MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS, POSITIVE LOGIC, FOUR-BIT FULL ADDER, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, CMOS logic microcircuits. Two product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Four-bit full adder
51	Four-bit full adder

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8", dual-in-line pack)
F	F-5 (16-lead, 1/4" x 3/8", flat pack)
Z	F-5 (16-lead, 1/4" x 3/8", flat pack, except A dimension = .1" (2.54 mm) max).

NOTES

1. As an exception to 3.5.6.2.3 of MIL-M-38510, for case outline Z only, the leads of bottom brazed ceramic packages (i.e. configuration 2 of case outline F-5) may have electroless nickel undercoating which shall be 50 to 200 microinches (1.27 to 5.08 µm) thick provided the lead finish is hot solder dip (i.e. finish letter A) and provided that, after any lead forming, an additional hot solder dip coating is applied which shall extend from the outer tip of the lead to no more than 0.015 inch (0.38 mm) from the package edge.
2. For bottom or side brazed packages, case outline Z only, the S_1 dimension may go to .000 inch (.00 mm) minimum.

1.3 Absolute maximum ratings.

Supply voltage range (V_{DD} - V_{SS}):
  Device type 01 - - - - - - - - - - - - - -0.5 V to +15.5 V
  Device type 51 - - - - - - - - - - - - - -0.5 V to +18.0 V
Input current (each input) - - - - - - - ±10 mA
Input voltage range- - - - - - - - - - - (V_{SS}-0.5 V) $\leq V_I \leq$ (V_{DD} + 0.5 V)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: George C. Marshall Space Flight Center, National Aeronautics and Space Administration, ATTN: EG02, Marshall Space Flight Center, AL 35812, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Storage temperature range	-65°C to +175°C
Maximum power dissipation, P_D	200 mW
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction to case	(See MIL-M-38510, appendix C)
Junction temperature	T_J = 175°C

1.4 Recommended operating conditions.

Supply voltage range (V_{DD} - V_{SS}):	
Device type 01	4.5 V dc to 12.5 V dc
Device type 51	4.5 V dc to 15.0 V dc
Input low (V_{IL}) voltage range:	
Device type 01	0 to 0.85 V dc at V_{DD} = 5 V; 0 to 2.1 V dc at V_{DD} = 12.5 V
Device type 51	0 to 1.5 V dc at V_{DD} = 5.0 V, V_{OL} = 10 percent V_{DD}, V_{OH} = 90 percent V_{DD}; 0 to 4.0 V dc at V_{DD} = 15 V; 0 to 2.0 V dc at V_{DD} = 10 V
Input high (V_{IH}) voltage range:	
Device type 01	3.95 to 5.0 V dc at V_{DD} = 5 V; 10.0 to 12.5 V dc at V_{DD} = 12.5 V
Device type 51	3.5 to 5.0 V dc at V_{DD} = 5.0 V, V_{OL} = 10 percent V_{DD}, V_{OH} = 90 percent V_{DD}; 11.0 to 15.0 V dc at V_{DD} = 15 V; 8.0 to 10.0 V dc at V_{DD} = 10 V
Load capacitance	50 pF maximum
Ambient operating temperature range	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a confliuct between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

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3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein. Although eutectic die bonding is preferred, epoxy die bonding may be performed. However, the resin used shall be Dupont 5504 Conductive Silver Paste, which is cured at 200° +10°C for a minimum of 2 hours. The use of equivalent epoxies or cure cycles shall be approved by the qualifying activity. Equivalency shall be demonstrated in data submitted to the qualifying activity for verification.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Logic diagrams. The logic diagrams shall be as specified on figure 2.

3.2.3 Truth tables. The truth tables shall be as specified on figure 3.

3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III. V_{IL} and V_{IH} testing requires only a summary of attributes data.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1 Total dose radiation hardness identifier. Total dose radiation hardness identifier shall be in accordance with MIL-M-38510 and 4.5.5 herein.

3.6.2 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.3 Correctness of indexing and markings. All devices shall be subjected to the final electrical tests specified in table II after part number marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 39 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Delete the sequence specified in 3.1.9 through 3.1.13 of method 5004 and substitute sequentially lines 1 through 7 of table II herein.
b. Burn-in tests (method 1015 of MIL-STD-883).
   1. Static tests (test condition A) use circuit shown on figure 4, or equivalent. Ambient temperature (T_A) shall be 125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I method 1015 for class B devices.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ V_{SS} = 0 V, -55°C ≤ T_A ≤ 125°C, Unless otherwise specified	Device type	Limits Min	Limits Max	Unit
Positive clamping input to V_{DD}	$V_{IC(POS)}$	T_A = 25°C, V_{DD} = GND, V_{SS} = Open, Output = Open, I_I = 1 mA	All		1.5	V
Negative clamping input to V_{SS}	$V_{IC(NEG)}$	T_A = 25°C, V_{DD} = Open, V_{SS} = GND, Output = Open, I_I = -1 mA	All		-6	V
Quiescent supply current	I_{SS}	Any combination of inputs, V_{DD}=15.0 V	01		-5	µA
		Any combination of inputs, V_{DD}=18.0 V	51		-5	µA
High level output voltage	V_{OH1}	V_{DD} = 5.0 V, I_{OH} = -0.05 mA (see table III)	01	4.5		V
	V_{OH2}	V_{DD} = 5.0 V, I_{OH} = 0 (see table III)	01	4.95		V
	V_{OH3}	V_{DD} = 12.5 V, I_{OH} = 0 (see table III)	01	11.25		V
	V_{OH4}	V_{DD} = 15.0 V, I_{OH} = 0 (see table III)	51	14.95		V
Low level output voltage	V_{OL1}	V_{DD} = 5.0 V, I_{OL} = 0.05 mA (see table III)	01		0.50	V
	V_{OL2}	V_{DD} = 5.0 V, I_{OL} = 0 (see table III)	01		0.05	V
	V_{OL3}	V_{DD} = 12.5 V, I_{OL} = 0 (see table III)	01		1.25	V
	V_{OL4}	V_{DD} = 15.0 V, I_{OL} = 0 (see table III)	51		0.05	V
Input high voltage	V_{IH1}	V_{DD} = 5.0 V, V_O = 0.5 V, $\|I_O\| \leq 1$ µA	51	3.5		V
	V_{IH2}	V_{DD} = 10.0 V, V_O = 1.0 V, $\|I_O\| \leq 1$ µA	51	7.0		V
	V_{IH3}	V_{DD} = 15.0 V, V_O = 1.5 V, $\|I_O\| \leq 1$ µA	51	11.0		V
Input low voltage	V_{IL1}	V_{DD} = 5.0 V, V_O = 4.5 V, $\|I_O\| \leq 1$ µA	51		1.5	V
	V_{IL2}	V_{DD} = 10.0 V, V_O = 9.0 V, $\|I_O\| \leq 1$ µA	51		3.0	V
	V_{IL3}	V_{DD} = 15.0 V, V_O = 13.5 V, $\|I_O\| \leq 1$ µA	51		4.0	V
Output low (sink) current	I_{OL1}	V_{DD} = 5.0 V, V_{IN} = V_{SS}, V_{OL} = 0.4 V	51	0.36		mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics -Continued.

Test	Symbol	Conditions 1/ V_{SS} = 0 V, -55°C ≤ T_A ≤ 125°C, Unless otherwise specified	Device type	Limits Min	Limits Max	Unit
Output low (sink) current	I_{OL2}	V_{DD} = 15.0 V, V_{IN} = V_{SS}, V_{OL} = 1.5 V	51	2.4		mA
Output high (source) current	I_{OH1}	V_{DD} = 5.0 V, V_{IN} = V_{DD}, V_{OH} = 4.6 V	51	-0.36		mA
	I_{OH2}	V_{DD} = 15.0 V, V_{IN} = V_{DD}, V_{OH} = 13.5 V	51	-2.4		mA
Input leakage current 2/ (high)	I_{IH}	Measure inputs sequentially, V_{DD} = 15.0 V	01		45	nA
		V_{DD} = 18.0 V	51			
Input leakage current 2/ (low)	I_{IL}	Measure inputs sequentially, V_{DD} = 15.0 V	01		-45	nA
		V_{DD} = 18.0 V	51			
Input voltage	V_{ZAP}	Ci = 100 pF, R2 = 1.5 kΩ (see 4.5.3)	All	400		V
Input capacitance	Ci	V_{DD} = 0 V, f = 1 MHz, T_A = 25°C	All		12	pF
Propagation delay high to low level - Sum input to sum output	t_{PHL1}	V_{DD} = 5.0 V, C_L = 50 pF (see figure 8)	All		3375	ns
Propagation delay high to low level - Carry input to sum output	t_{PHL2}		All		3375	ns
Propagation delay high to low level - Sum input to carry output	t_{PHL3}		All		1125	ns
Propagation delay high to low level - Carry input to carry output	t_{PHL4}		All		450	ns
Propagation delay low to high level - Sum input to sum output	t_{PLH1}		All		4350	ns
Propagation delay low to high level - Carry input to sum output	t_{PLH2}		All		4350	ns
Propagation delay low to high level - Sum input to carry output	t_{PLH3}		All		1125	ns
Propagation delay low to high level - Carry input to carry output	t_{PLH4}		All	31	450	ns
Transition times at sum outputs	t_{THL}, t_{TLH}		All		15000	ns
Transition times at carry outputs	t_{THL}, t_{TLH}		All		780	ns

1/ Complete terminal conditions shall be as specified in table III.
2/ Input current at one input node.

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TABLE II. Burn-in and electrical test requirements.

Line no.	Applicable tests and MIL-STD-883 test methods	Class S device 3/				Class B device 3/			
		Ref. par.	Table 2/ III sub-groups	Table 1/ IV delta limits	Test circuit figure	Ref. par.	Table 2/ III sub-groups	Table 1/ IV delta limits	Test circuit figure
1	Interim electrical parameters (method 5004)		1				1		
2	Static burn-in I (method 1015)	4.2b. 4.5.2			4				
3	Same as line 1		1	Δ					
4	Static burn-in II (method 1015)	4.2b. 4.5.2			4	4.2b. 4.5.2	4/		4
5	Same as line 1	4.2d.	1*	Δ		4.2d.	1*	Δ	
6	Dynamic burn-in (method 1015)	4.2b. 4.5.2			5				
7	Same as line 1	4.2d.	1*	Δ					
8	Final electrical parameters (method 5004)		1*,2,3,7, 9				1*,2,3,7, 9		
9	Group A end point electrical parameters (method 5005)	4.4.1	1,2,3,4, 7,8,9, 10,11			4.4.1	1,2,3, 4,7,9, 10,11		
10	Group B end point electrical parameters (method 5005)	4.4.2	1,2,3,7, 8,9,10,11	Δ					
11	Group C end-point electrical parameters (method 5005)					4.4.3c.	1,2,3	Δ	
12	Additional group C tests (method 5005)					4.4.3c.	10,11		
13	Group D end-point electrical parameters (method 5005)	4.4.4	1,2,3			4.4.4	1,2,3		

1/ (Δ) indicates delta limits shall be required on table III subgroup 1, where specified, and delta values shall be computed with reference to the previous interim electrical parameters.
2/ (*) indicates PDA applies to subgroup 1 (see 4.2.1).
3/ Blank spaces indicate tests are not applicable.
4/ The device manufacturer may, at his option, either perform delta measurements or within 24 hours after burn-in (or removal of bias) perform the final electrical parameter measurements.

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Cases E, F and Z

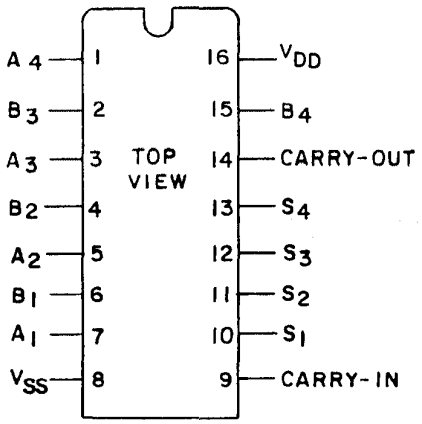


FIGURE 1. Terminal connections.

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Device types 01 and 51

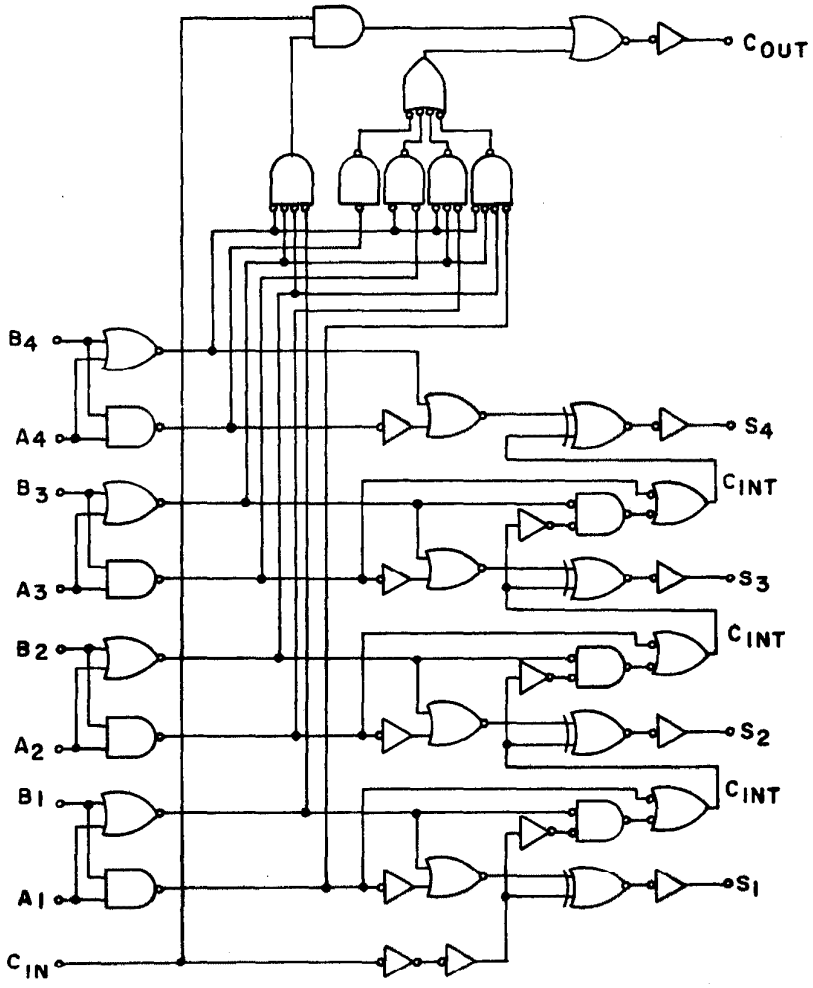


FIGURE 2. Logic diagram.

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A_n	B_n	C_{in}	C_{int}*	C_{OUT}	SUM
L	L	L	L	L	L
H	L	L	L	L	H
L	H	L	L	L	H
H	H	L	H	L(H**)	L
L	L	H	L	L	H
H	L	H	H	L(H**)	L
L	H	H	H	L(H**)	L
H	H	H	H	L(H**)	H

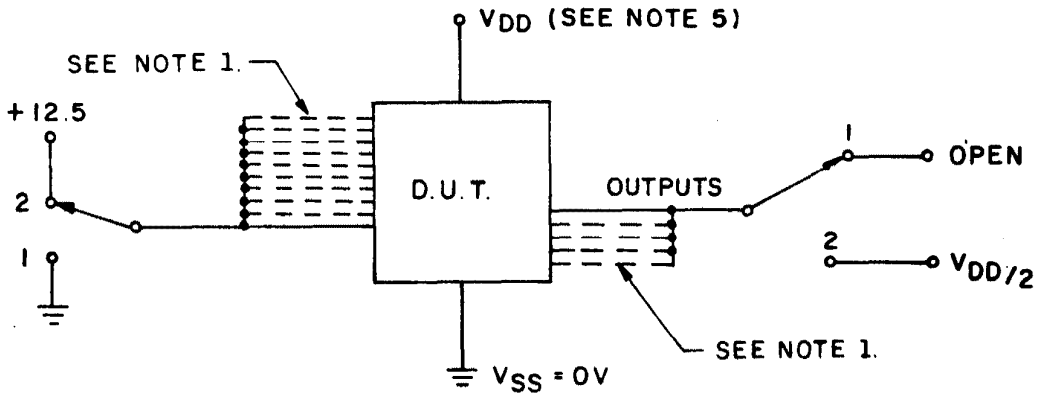
* = Internal Carry, ** C_{OUT} = I when N = 4

L = Low
H = High

Positive logic one stage

FIGURE 3. Truth table.

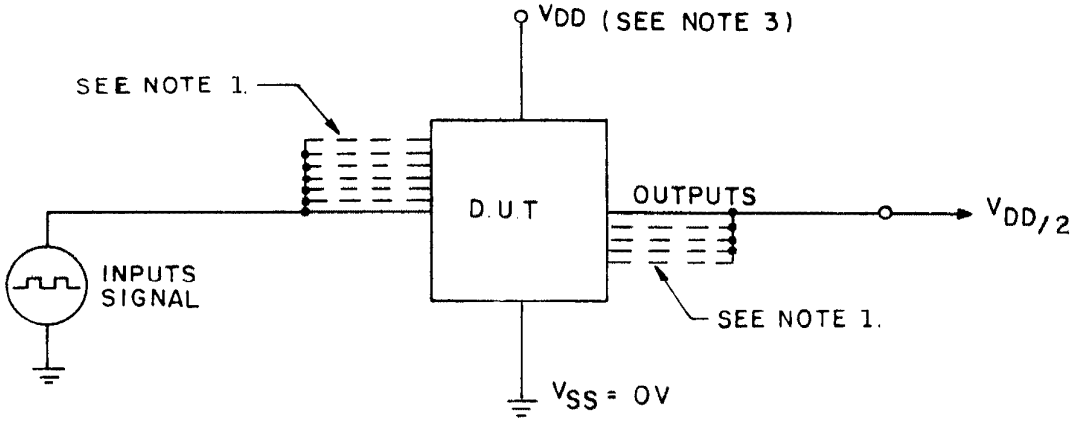
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NOTES:
1. For static burn-in I, all inputs are connected to 0 volts, switch position 1.
2. For static burn-in II, all inputs are connected to V_{DD}, switch position 2.
3. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exeed ±20% of its branded value due to use, heat or age.
4. Output may be in switch position 1 or 2.
5. V_{DD} = 12.5 V minimum, 15.0 V maximum for device types 01.
   V_{DD} = 15.0 V minimum, 18.0 V maximum for device types 51.
   $V_{DD/2} = V_{DD}/2$ ±1.0 V for all devices.
   V_{SS} = 0.0 V.

FIGURE 4. Static burn-in test circuits.

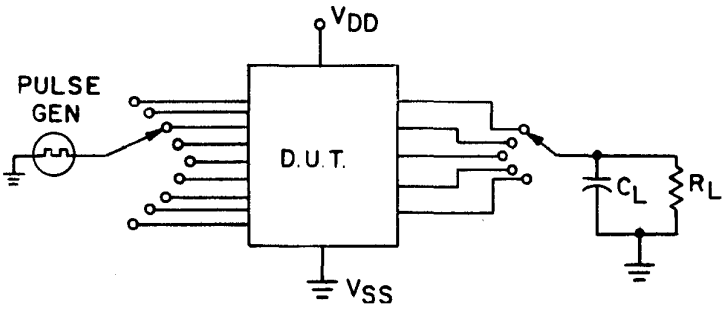
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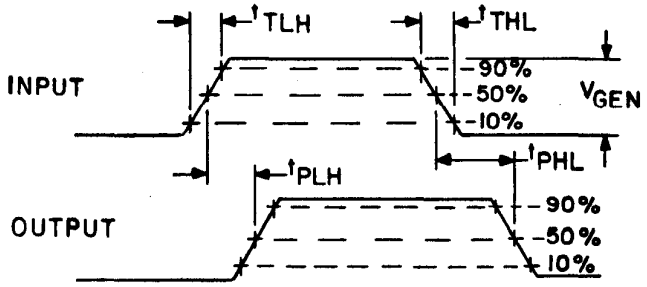
NOTES:
1. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.
2. Input signal requirements:
   a. Square wave, 50% duty cycle.
   b. 25 kHz < PRR < 1 MHz.
   c. t_{TLH} and t_{THL} < 1 µs.
   d. Voltage level:
      Minimum = V_{SS} -0.5 V, +10% V_{DD}.
      Maximum = V_{DD} +0.5, -10% V_{DD}.
3. V_{DD} = 12.5 V minimum, 15.0 V maximum for device types 01.
   V_{DD} = 15 V minimum, 18 V maximum for device types 51.
   $V_{DD/2}$ = $V_{DD}/2$ ±1.0 V.
   V_{SS} = 0.0 V.

FIGURE 5. Dynamic burn-in and steady state life test circuit.

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R_L = 200 kΩ ±10%.
C_L = 50 pF ±5.0 (includes wiring and probe capacitance)



NOTES:
1. The pulse generator has the following characteristics: $V_{gen} = V_{DD}$ ±1%, duty cycle = 50%, $t_{TLH} = t_{THL}$ = 20 ±2.0 ns and pulse repetition period = 50 ±5 µs.
2. See table III for complete terminal conditions.

FIGURE 6. Switching time test circuit and waveforms.

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TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions and limits 1/																Measured terminal	Test limits						Units
																				Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test No.	A4	B3	A3	B2	A2	B1	A1	V_{SS}	C_{IN}	S1	S2	S3	S4	C_{OUT}	B4	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(POS)}$		1	1 mA															GND	A4		1.5					Vdc
		2		1 mA															B3							
		3			1 mA														A3							
		4				1 mA													B2							
		5					1 mA												A2							
		6						1 mA											B1							
		7							1 mA										A1							
		8									1 mA								C_{IN}							
		9															1 mA		B4							
$V_{IC(NEG)}$		10	-1 mA							GND									A4		-6					Vdc
		11		-1 mA															B3							
		12			-1 mA														A3							
		13				-1 mA													B2							
		14					-1 mA												A2							
		15						-1 mA											B1							
		16							-1 mA										A1							
		17									-1 mA								C_{IN}							
		18															-1 mA		B4							
I_{SS}	3005	19	GND	GND	GND	GND	GND	GND	GND		GND						GND	15 V	V_{SS}		-0.5		-5.0			μA
		20	15 V	15 V	15 V	15 V	15 V	GND	GND		15 V						15 V									
		21	15 V		GND	GND	GND	15 V	15 V								15 V									
		22	15 V		GND	15 V	15 V	15 V	GND								GND									
		23	GND		15 V		15 V	15 V	15 V								GND									
		24	15 V		GND		GND	GND	GND								GND									
		25	GND		15 V		GND	15 V	GND		GND						15 V									
		26	15 V	GND	GND		15 V	15 V	GND		15 V						GND									
		27	15 V	15 V	GND		GND	GND	15 V		15 V						GND									
		28	15 V	15 V	GND	GND	GND	15 V			15 V						GND									
		29	GND	GND	15 V	GND	15 V	GND			GND						15 V									
		30	GND	GND	GND	GND		15 V			15 V						15 V									
		31	15 V	15 V	GND	15 V		GND	GND		15 V						GND									
		32		GND	15 V	15 V		15 V	GND		15 V															
		33		GND	15 V	15 V	GND	15 V	15 V		GND															
		34		15 V	GND	15 V	GND	15 V	GND		15 V															
		35			15 V	GND	GND	GND			GND															
		36			GND	GND	15 V	15 V			15 V															
		37	GND		GND	15 V	15 V	15 V			15 V						15 V									
		38	15 V		GND	GND	15 V	15 V	15 V		GND						GND									
		39	GND	GND	15 V	GND	15 V	15 V	GND		15 V						15 V									
		40	15 V			GND	GND	GND	15 V		15 V						GND									
		41				GND	15 V	15 V	GND		GND															
		42				15 V	GND	15 V	GND		GND															
		43				15 V	GND	GND	15 V		GND															
		44				GND	15 V		GND		15 V															
		45	GND	15 V	GND	GND	15 V		15 V		15 V						15 V									
		46	GND	15 V	GND	GND	GND		GND		GND						15 V									

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions and limits 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test No.	A4	B3	A3	B2	A2	B1	A1	V_{SS}	C_{IN}	S1	S2	S3	S4	C_{OUT}	B4	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{OH1}	3006	47	V_{IL1} 2/	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	V_{IH1}	GND	V_{IL1}	I_{OH1} 4/	I_{OH1}	I_{OH1}	I_{OH1}		V_{IL1}	5.0 V	S1	4.5		4.5		4.5		Vdc
		48			V_{IL1}		V_{IH1}		V_{IL1}										S2							
		49			V_{IH1}		V_{IL1}												S3							
		50	V_{IH1} 3/		V_{IL1}											I_{OH2} 5/	V_{IH1}		S4							
		51			V_{IL1}											I_{OH2}	V_{IL1}		C_{OUT}							
		52			V_{IH1}		V_{IH1}		V_{IH1}		V_{IH1}								C_{OUT}							
V_{OH2}		53	V_{IL1}		V_{IL1}		V_{IL1}		V_{IH1}		V_{IL1}								S1	4.95		4.95		4.95		
		54			V_{IL1}		V_{IH1}		V_{IL1}										S2							
		55			V_{IH1}		V_{IL1}												S3							
		56	V_{IH1}		V_{IL1}														S4							
		57			V_{IL1}												V_{IH1}		C_{OUT}							
		58			V_{IH1}		V_{IH1}		V_{IH1}		V_{IH1}						V_{IL1}		C_{OUT}							
V_{OH3}		59	V_{IL2} 6/	V_{IL2}	V_{IL2}	V_{IL2}	V_{IL2}	V_{IL2}	V_{IH2}		V_{IL2}						V_{IL2}	12.5 V	S1	11.25		11.25		11.25		
		60						V_{IH2}	V_{IL2}		V_{IL2}								S1							
		61						V_{IL2}	V_{IL2}		V_{IH2}								S1							
		62						V_{IH2}	V_{IH2}		V_{IH2}								S1							
		63				V_{IH2}		V_{IH2}	V_{IH2}		V_{IL2}								S2							
		64				V_{IH2}	V_{IH2}	V_{IH2}	V_{IH2}										S2							
		65				V_{IL1}	V_{IL2}	V_{IH2}	V_{IH2}										S2							
		66				V_{IL2}	V_{IH2}	V_{IL2}	V_{IH2}										S2							
		67		V_{IH2}		V_{IL2}	V_{IL2}												S3							
		68		V_{IL2}	V_{IH2}	V_{IL2}	V_{IL2}												S3							
		69		V_{IH2}	V_{IH2}	V_{IH2}	V_{IH2}												S3							
		70		V_{IL2}	V_{IL2}	V_{IH2}	V_{IH2}												S3							
		71	V_{IH2} 7/			V_{IL2}	V_{IL2}												S4							
		72	V_{IL2}	V_{IH2}	V_{IH2}												V_{IH2}		S4							
		73	V_{IH2}	V_{IH2}	V_{IH2}												V_{IH2}		S4							
		74	V_{IL2}	V_{IH2}	V_{IH2}												V_{IL2}		S4							
		75	V_{IH2}	V_{IL2}	V_{IH2}		V_{IH2}		V_{IH2}		V_{IH2}						V_{IL2}		C_{OUT}							
		76	V_{IL2}	V_{IH2}	V_{IL2}	V_{IH2}	V_{IL2}	V_{IH2}	V_{IL2}		V_{IH2}						V_{IH2}		C_{OUT}							
		77	V_{IH2}	V_{IL2}	V_{IL2}	V_{IL2}	V_{IL2}	V_{IL2}	V_{IL2}		V_{IL2}						V_{IH2}		C_{OUT}							
V_{OL1}	3007	78	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}	V_{IL1}		V_{IL1}	I_{OL1} 8/	I_{OL1}	I_{OL1}	I_{OL1}		V_{IL1}	5.0 V	S1		0.5		0.5		0.5	Vdc
		79																	S2							
		80																	S3							
		81																	S4							
		82														I_{OL2} 9/			C_{OUT}							
V_{OL2}		83																	S1		.05		.05		.05	Vdc
		84																	S2							
		85																	S3							
		86																	S4							
		87																	C_{OUT}							

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions and limits 1/															Measured terminal	Test limits						Units
																			Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16							
		Test No.	A4	B3	A3	B2	A2	B1	A1	V_{SS}	C_{IN}	S1	S2	S3	S4	C_{OUT}	B4	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{OL3}	3007	88	V_{IL2}	V_{IL2}	V_{IL2}	V_{IL2}	V_{IL2}	V_{IL2}	V_{IL2}	GND	V_{IL2}						V_{IL2}	12.5 V	S1		1.25		1.25		1.25	Vdc
		89						V_{IL2}	V_{IH2}		V_{IH2}								S1							
		90						V_{IH2}	V_{IL2}		V_{IH2}								S1							
		91						V_{IH2}	V_{IH2}		V_{IL2}								S1							
		92				V_{IH2}	V_{IH2}	V_{IL2}	V_{IL2}										S2							
		93				V_{IH2}	V_{IL2}	V_{IH2}	V_{IL2}										S2							
		94				V_{IL2}	V_{IH2}	V_{IL2}	V_{IH2}										S2							
		95					V_{IL2}	V_{IH2}	V_{IL2}										S2							
		96					V_{IL2}	V_{IL2}											S3							
		97		V_{IH2}	V_{IH2}		V_{IL2}												S3							
		98		V_{IH2}	V_{IL2}	V_{IH2}	V_{IH2}												S3							
		99		V_{IL2}	V_{IH2}	V_{IH2}	V_{IH2}												S3							
		100				V_{IL2}	V_{IL2}												S4							
		101	V_{IH2}														V_{IH2}		S4							
		102	V_{IH2}	V_{IH2}													V_{IL2}		S4							
		103	V_{IL2}	V_{IH2}													V_{IH2}		S4							
		104	V_{IL2}	V_{IL2}	V_{IL2}												V_{IL2}		C_{OUT}							
I_{IH1} 10/	3010	105	15 V	15 V	15 V	15 V	15 V	15 V	15 V		15 V						15 V	15 V	All inputs together		9.0					nA
I_{IH2}	3010	106	15 V	GND	GND	GND	GND	GND	GND		GND						GND		A4		1.0		45			nA
		107	GND	15 V	GND														B3							
		108		GND	15 V														A3							
		109			GND	15 V													B2							
		110				GND	15 V												A2							
		111					GND	15 V											B1							
		112						GND	15 V										A1							
		113							GND		15 V								C_{IN}							
		114									GND						15 V		B4							
I_{IL1} 10/	3009	115															GND	15 V	All inputs together		-9.0					nA
I_{IL2}	3009	116																	A4		-1.0		-45			nA
		117																	B3							
		118																	A3							
		119																	B2							
		120																	A2							
		121																	B1							
		122																	A1							
		123																	C_{IN}							
		124																	B4							

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions and limits 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 4 T_A = 25°C						
		Test No.	A4	B3	A3	B2	A2	B1	A1	V_{SS}	C_{IN}	S1	S2	S3	S4	C_{OUT}	B4	V_{DD}		Min	Max					
Ci	3012	125	11/							GND								GND	A4		15					pF
		126		11/															B3								
		127			11/														A3								
		128				11/													B2								
		129					11/												A2								
		130						11/											B1								
		131							11/										A1								
		132									11/								C_{IN}								
		133															11/		B4								
																				Subgroup 7 T_A = 25°C		Subgroup 8 T_A=125° C		T_A=-55° C			
																				Min	Max	Min	Max	Min	Max		
Truth table test	3014	134	GND	GND	GND	GND	GND	GND	GND	GND	GND	L	L	L	L	L	GND	5 V	All outputs								
		135	5 V	5 V	5 V	5 V	5 V	GND	GND		5 V	H	L	H	H	H	5 V										
		136	5 V		GND	GND	GND	5 V	5 V			H	H	H	L	H	5 V										
		137	5 V		GND	5 V	5 V	5 V	GND			L		L	L	H	GND										
		138	GND		5 V		5 V	5 V	5 V			H		H	H	L	GND										
		139	5 V		GND		GND	GND	GND			H		H	H	L	GND										
		140	GND		5 V		GND	5 V	GND		GND	H		L	L	H	5 V										
		141	5 V	GND	GND		5 V	5 V	GND		5 V	L		H	H	L	GND										
		142	5 V	5 V	GND		GND	GND	5 V		5 V	L		L	L	H	GND										
		143	5 V	5 V	GND	GND	GND	5 V			5 V	H		H	H	L	GND										
		144	GND	GND	5 V	GND	5 V	GND			GND			H	H	L	5 V										
		145	GND	GND	GND	GND		5 V			5 V			L	H	L	5 V										
		146	5 V	5 V	GND	5 V		GND	GND		5 V			L	L	H	GND										
		147		GND	5 V			5 V	GND		5 V	L		H								12/, 13/					
		148		GND	5 V		GND	5 V	5 V		GND			L													
		149		5 V	GND		GND	5 V	GND		5 V																
		150			5 V	GND	GND	GND			GND																
		151			GND	GND	5 V	5 V			5 V																
		152	GND		GND	5 V					5 V			H			5 V										
		153	5 V		GND	GND			5 V		GND			L			GND										
		154	GND	GND	5 V				GND		5 V			L			5 V										
		155	5 V	GND	5 V		GND	GND	5 V		5 V			H	H	L	GND										
		156					5 V	5 V	GND		GND	H															
		157				5 V	GND	5 V	GND		GND																
		158				5 V	GND	GND	5 V		GND																
		159				GND	5 V		GND		5 V																
		160	GND	5 V	GND	GND	5 V		5 V		5 V	L	L	L	L	H	5 V										
		161	GND	5 V	GND	GND	GND		GND		GND	L	L	H	H	L	5 V										

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions and limits 1/																Measured terminal	Test limits						Units
																				Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test No.	A4	B3	A3	B2	A2	B1	A1	V_{SS}	C_{IN}	S1	S2	S3	S4	C_{OUT}	B4	V_{DD}		Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 (Fig 8)	162	GND	GND	GND	GND	GND	GND	IN	GND	GND	OUT					GND	5 V	A1 to S1		2250		3375		2250	ns
		163					GND	IN	GND			OUT							B1 to S1							
		164					IN	GND					OUT						A2 to S2							
		165				IN	GND						OUT						B2 to S2							
		166			IN	GND								OUT					A3 to S3							
		167		IN	GND									OUT					B3 to S3							
		168		GND											OUT				A4 to S4							
		169	IN												OUT		IN		B4 to S4							
t_{PHL2}		170							5 V		IN	OUT					GND		C_{IN} to S1							
		171							5 V				OUT						C_{IN} to S2							
		172					5 V		5 V					OUT					C_{IN} to S3							
		173			5 V		5 V		5 V						OUT				C_{IN} to S4							
t_{PHL3}		174	5 V		GND		GND		GND		GND					OUT	IN		B4 to C_{OUT}		750		1125		750	
		175	IN		GND		GND		GND		GND						5 V		A4 to C_{OUT}		750		1125		750	
t_{PHL4}		176	5 V		5 V		5 V		5 V		IN						GND		C_{IN} to C_{OUT}		300		450		300	
		177	GND	5 V	GND	5 V	GND	5 V	GND		IN						5 V		C_{IN} to C_{OUT}		300		450		300	
t_{PLH1}		178	GND	GND	GND	GND	GND	GND	IN		GND	OUT					GND		A1 to S1		2900		4350		2900	
		179					GND	IN	GND			OUT							B1 to S1							
		180					IN	GND					OUT						A2 to S2							
		181				IN	GND						OUT						B2 to S2							
		182			IN	GND								OUT					A3 to S3							
		183		IN	GND									OUT					B3 to S3							
		184		GND											OUT				A4 to S4							
		185	IN												OUT		IN		B4 to S4							
t_{PLH2}		186							5 V		IN	OUT					GND		C_{IN} to S1							
		187							5 V				OUT						C_{IN} to S2							
		188					5 V		5 V					OUT					C_{IN} to S3							
		189			5 V		5 V		5 V						OUT				C_{IN} to S4							
t_{PLH3}		190	5 V		GND		GND		GND		GND					OUT	IN		B4 to C_{OUT}		750		1125		750	
		191	IN		GND		GND		GND		GND						5 V		A4 to C_{OUT}		750		1125		750	
t_{PLH4}		192	5 V		5 V		5 V		5 V		IN						GND		C_{IN} to C_{OUT}		300		450		300	
		193	GND	5 V	GND	5 V	GND	5 V	GND		IN						5 V		C_{IN} to C_{OUT}		300		450		300	

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E, F, Z	Terminal conditions and limits 1/															Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Test No.	A4	B3	A3	B2	A2	B1	A1	V_{SS}	C_{IN}	S1	S2	S3	S4	C_{OUT}	B4	V_{DD}		Min	Max	Min	Max	Min	Max	
t_{THL}	3004 (Fig 8)	194	GND	GND	GND	GND	GND	GND	IN	GND	GND	OUT	OUT	OUT	OUT		GND	5 V	S1		10000		15000		10000	ns
		195		GND	GND	IN	5 V	5 V	5 V										S2							
		196		5 V	IN	5 V	5 V	GND	GND										S3							
		197	IN	5 V	5 V	GND	GND										5 V		S4							
		198	IN	GND	GND	GND	GND									OUT	5 V		C_{OUT}		520		780		520	
		199	5 V	GND	5 V	GND	5 V		5 V		IN					OUT	GND		C_{OUT}		520		780		520	
t_{TLH}		200	GND	GND	GND	GND	GND	GND	IN		GND	OUT	OUT	OUT	OUT		GND		S1		10000		15000		10000	
		201		GND	GND	IN	5 V	5 V	5 V										S2							
		202		5 V	IN	5 V	5 V	GND	GND										S3							
		203	IN	5 V	5 V	GND	GND										5 V		S4							
		204	IN	GND	GND	GND	GND									OUT	5 V		C_{OUT}		520		780		520	
		205	5 V	GND	5 V	GND	5 V		5 V		IN					OUT	GND		C_{OUT}		520		780		520	

1/ Pins not designated can be "high" level logic, "low" level logic, or open. Exceptions are as follows: $V_{IC(POS)}$ tests, the V_{SS} terminal shall be open; $V_{IC(NEG)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.
2/ V_{IL1} = 1.1 V @ 25°C: 0.85 V @ 125°C: 1.35 V @ -55°C.
3/ V_{IH1} = 3.8 V @ 25°C: 3.6 V @ 125°C: 3.95 V @ -55°C.
4/ I_{OH1} = -0.05 mA @ 25°C: -0.35 mA @ 125°C: -0.06 mA @ -55°C.
5/ I_{OH2} = -0.25 mA @ 25°C: -0.175 mA @ 125°C: -0.31 mA @ -55°C.
6/ V_{IL2} = 2.8 V @ 25°C: 2.55 V @ 125°C: 3.05 V @ -55°C.
7/ V_{IH2} = 9.5 V @ 25°C: 9.25 V @ 125°C: 9.75 V @ -55°C.
8/ I_{OL1} = 0.05 mA @ 25°C: 0.035 mA @ 125°C: 0.06 mA @ -55°C.
9/ I_{OL2} = 0.25 mA @ 25°C: 0.175 mA @ 125°C: 0.31 mA @ -55°C.
10/ The device manufacturer may, at his optior, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all input or measure all inputs together.
11/ See 4.4.1c.
12/ The truth table tests shall be performed in sequence.
13/ The truth table tests shall be performed at V_{IH} and $V_{DD} \leq 5.0$ V and ≥ 15.0 V. L = V_{SS} + 0.50 V maximum and H = V_{DD} - 0.50 V minimum.

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TABLE III. Group A inspection for device type 51.

Symbol	MIL-STD-883 method	Cases E,F,Z Symbol	Terminal conditions 1/ A_4	B_3	A_3	B_2	A_2	B_1	A_1	V_{SS}	C_{IN}	S_1	S_2	S_3	S_4	C_{OUT}	B_4	V_{DD}	Measured terminal	Test limits Subgroup 1 T_A=25°C Min	Subgroup 1 T_A=25°C Max	Subgroup 2 T_A=125°C Min	Subgroup 2 T_A=125°C Max	Subgroup 3 T_A=-55°C Min	Subgroup 3 T_A=-55°C Max	Unit
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
$V_{IC(POS)}$		1	1 mA															GND	A_4		1.5					Vdc
		2		1 mA															B_3							
		3			1 mA														A_3							
		4				1 mA													B_2							
		5					1 mA												A_2							
		6						1 mA											B_1							
		7							1 mA										A_1							
		8									1 mA								C_{IN}							
		9															1 mA		B_4							
$V_{IC(NEG)}$		10	-1 mA							GND									A_4		-.5					
		11		-1 mA															B_3							
		12			-1 mA														A_3							
		13				-1 mA													B_2							
		14					-1 mA												A_2							
		15						-1 mA											B_1							
		16							-1 mA										A_1							
		17									-1 mA								C_{IN}							
		18															-1 mA		B_4							
I_{SS} 2/	3005	19	GND	GND	GND	GND	GND	GND	GND		GND						GND	18.0 V	V_{SS}		-0.5		-5.0			μA
		20	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	GND	GND		18.0 V						18.0 V									
		21	18.0 V		GND	GND	GND	18.0 V	18.0 V								18.0 V									
		22	18.0 V		GND	18.0 V	18.0 V	18.0 V	GND								GND									
		23	GND		18.0 V		GND	GND	GND								GND									
		24	18.0 V		GND		GND	18.0 V	GND		GND						GND									
		25	GND		18.0 V		GND	18.0 V	GND		18.0 V						18.0 V									
		26	18.0 V	GND	GND		18.0 V	GND	18.0 V		18.0 V						GND									
		27	18.0 V	18.0 V	GND		GND	GND	18.0 V		18.0 V						GND									
		28	18.0 V	18.0 V	GND	GND	GND	18.0 V			18.0 V						GND									
		29	GND	GND	18.0 V	GND	18.0 V	GND			GND						18.0 V									
		30	GND	GND	GND	GND		18.0 V			18.0 V						18.0 V									
		31	18.0 V	18.0 V	GND	18.0 V		GND	GND		18.0 V						GND									
		32		GND	18.0 V			18.0 V	GND		18.0 V															
		33		GND	18.0 V		GND	18.0 V	18.0 V		GND															
		34		18.0 V	GND		GND	18.0 V	GND		18.0 V															
		35			18.0 V	GND	GND	GND			GND															
		36			GND	GND	18.0 V	18.0 V			18.0 V															
		37	GND		GND	18.0 V			18.0 V		18.0 V						18.0 V									
		38	18.0 V		GND	GND			GND		GND						GND									
		39	GND	GND	18.0 V				GND		18.0 V						18.0 V									
		40	18.0 V				GND	GND	18.0 V		18.0 V						GND									
		41					18.0 V	18.0 V	GND		GND															
		42				18.0 V	GND	GND	18.0 V		GND															
		43				GND	18.0 V		GND		18.0 V															
		44	GND	18.0 V	GND		18.0 V				18.0 V						18.0 V									
		45		18.0 V			GND				GND						18.0 V									
V_{OH4}	3006	47		GND				15.0 V	15.0 V		GND						GND	15.0 V	S_1	14.95		14.95		14.95		Vdc
		48						GND	GND		GND															
		49									15.0 V															
		50						15.0 V	15.0 V		15.0 V															
		51						15.0 V	15.0 V		GND								S_2							
		52				15.0 V	15.0 V	15.0 V	15.0 V																	
		53				GND	15.0 V	GND	GND										S_3							
		54				15.0 V	GND																			
		55		15.0 V		GND	GND												S_4							
		56		GND	15.0 V	15.0 V	GND																			
		57		15.0 V	15.0 V	15.0 V	15.0 V																			
		58		GND	GND	15.0 V	15.0 V																			
		59	15.0 V	GND	GND	GND											15.0 V									
		60	GND	GND	GND												15.0 V									
		61	15.0 V	15.0 V	15.0 V												GND									
		62	GND	15.0 V	15.0 V												GND		C_{OUT}							
		63	15.0 V	GND	15.0 V		15.0 V		15.0 V		15.0 V						15.0 V		C_{OUT}							
		64	GND	15.0 V	GND	15.0 V	GND	15.0 V	GND		15.0 V						15.0 V		C_{OUT}							
		65	15.0 V	GND	GND	GND	GND	GND	GND		GND															

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TABLE III. Group A inspection for device type 51 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z Symbol	Terminal conditions 1/ A_4	B_3	A_3	B_2	A_2	B_1	A_1	V_{SS}	C_{IN}	S_1	S_2	S_3	S_4	C_{OUT}	B_4	V_{DD}	Measured terminal	Test limits Subgroup 1 T_A=25°C Min	Max	Subgroup 2 T_A=125°C Min	Max	Subgroup 3 T_A=-55°C Min	Max	Unit
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
V_{OL4}	3007	66	GND	GND	GND	GND	GND	GND	GND	GND	GND						GND	15.0 V	S_1		0.05		0.05		0.05	Vdc
		67						GND	15.0 V		15.0 V															
		68						15.0 V	GND		15.0 V								S_2							
		69						15.0 V	15.0 V		GND															
		70				15.0 V	15.0 V	GND	GND										S_3							
		71				15.0 V	GND	GND	GND																	
		72				GND	15.0 V	15.0 V	15.0 V										S_4							
		73				GND	15.0 V	15.0 V	15.0 V																	
		74				GND	GND	GND	GND																	
		75		15.0 V	15.0 V	GND	GND										15.0 V									
		76		15.0 V	GND	15.0 V	15.0 V										GND									
		77		GND	15.0 V	15.0 V	15.0 V										15.0 V									
		78		GND		GND	GND										GND		C_{OUT}							
		79	15.0 V	GND																						
		80	15.0 V	15.0 V																						
		81	GND	15.0 V																						
		82	GND	GND	GND																					
V_{IH1}		83	3/	3/	3/	3/	3/	3/	3/		3/	3/	3/	3/	3/	3/	3/	3/	All outputs	3/	3/	3/	3/	3/	3/	
V_{IH2}		84																								
V_{IH3}		85																								
V_{IL1}		86																								
V_{IL2}		87																								
V_{IL3}		88																								
I_{OL1}		89	GND	GND	GND	GND	GND	GND	GND		GND	0.4 V					GND	5.0 V	S_1	0.51		0.36		0.64		mA
		90											0.4 V						S_2							
		91												0.4 V					S_3							
		92													0.4 V				S_4							
		93														0.4 V			C_{OUT}							
I_{OL2}		94										1.5 V						15.0 V	S_1	3.4		2.4		4.2		
		95											1.5 V						S_2							
		96												1.5 V					S_3							
		97													1.5 V				S_4							
		98														1.5 V			C_{OUT}							
I_{OH1}		99	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V		5.0 V	4.6 V					5.0 V	5.0 V	S_1	-0.51		-0.36		-0.64		
		100											4.6 V						S_2							
		101												4.6 V					S_3							
		102													4.6 V				S_4							
		103														4.6 V			C_{OUT}							
I_{OH2}		104	15.0 V	15.0 V	15.0 V	15.0 V	15.0 V	15.0 V	15.0 V		15.0 V	13.5 V					15.0 V	15.0 V	S_1	-3.4		-2.4		-4.2		
		105											13.5 V						S_2							
		106												13.5 V					S_3							
		107													13.5 V				S_4							
		108														13.5 V			C_{OUT}							
I_{IH1} 2/	3010	109	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V		18.0 V						18.0 V	18.0 V	All inputs together		9					nA
I_{IH2}		110	18.0 V	GND	GND	GND	GND	GND	GND		GND						GND		A_4		1		45			
		111	GND	18.0 V	GND	GND													B_3							
		112		GND	18.0 V	GND													A_3							
		113			GND	18.0 V													B_2							
		114				GND	18.0 V												A_2							
		115					GND	18.0 V											B_1							
		116						GND	18.0 V										A_1							
		117						GND	GND		18.0 V								C_{IN}							
		118						GND	GND		GND						18.0 V		B_4							

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TABLE III. Group A inspection for device type 51 - Continued.

Symbol	MIL-STD-883 method	Cases E,F,Z Symbol	Terminal conditions 1/																Measured terminal	Test limits						Unit
			A_4	B_3	A_3	B_2	A_2	B_1	A_1	V_{SS}	C_{IN}	S_1	S_2	S_3	S_4	C_{OUT}	B_4	V_{DD}		Subgroup 1 T_A=25°C		Subgroup 2 T_A=125°C		Subgroup 3 T_A=-55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
I_{IL1} 4/	3009	119	GND	GND	GND	GND	GND	GND	GND	GND	GND						GND	18.0 V	All inputs together		-9					nA
I_{IL2}		120	GND	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V		18.0 V						18.0 V		A_4		-1		-45			
		121	18.0 V	GND	18.0 V	18.0 V													B_3							
		122		18.0 V	GND														A_3							
		123			18.0 V	GND													B_2							
		124				18.0 V	GND												A_2							
		125					18.0 V	GND											B_1							
		126						18.0 V	GND										A_1							
		127						18.0 V	18.0 V		GND								C_{IN}							
		128						18.0 V	18.0 V		18.0 V						GND		B_4							
																				Subgroup 4 T_A = 25°C						
																				Min	Max					
Ci	3012	129	5/							GND								GND	A_4		7.5					pF
		130		5/															B_3							
		131			5/														A_3							
		132				5/													B_2							
		133					5/												A_2							
		134						5/											B_1							
		135							5/										A_1							
		136									5/								C_{IN}							
		137															5/		B_4							
																				Subgroup 7 T_A = 25°C		Subgroup 8 T_A =125°C		T_A = -55°C		
																				Min	Max	Min	Max	Min	Max	
Truth table	3014	138	GND	GND	GND	GND	GND	GND	GND	GND	GND	L	L	L	L	L	GND	5.0 V	All outputs							6/,7/
		139	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	GND	GND		5.0 V	H	L	H	H	H	5.0 V									
		140	5.0 V		GND	GND	GND	5.0 V	5.0 V			H	H	H	L	H	5.0 V									
		141	5.0 V		GND	5.0 V	5.0 V	5.0 V	GND			L		L	L	H	GND									
		142	GND		5.0 V		5.0 V	5.0 V	5.0 V			H		H	H	L	GND									
		143	5.0 V		GND		GND	GND	GND			H		H	H	L	GND									
		144	GND		5.0 V		GND	5.0 V	GND		GND	H		L	L	H	5.0 V									
		145	5.0 V	GND	GND		5.0 V	5.0 V	GND		5.0 V	L		H	H	L	GND									
		146	5.0 V	5.0 V	GND		GND	GND	5.0 V		5.0 V	L		L	L	H	GND									
		147	5.0 V	5.0 V	GND	GND	GND	5.0 V			5.0 V	H		H	H	L	GND									
		148	GND	GND	5.0 V	GND	5.0 V	GND			GND			H	H	L	5.0 V									
		149	GND	GND	GND	GND		5.0 V			5.0 V			L	H	L	5.0 V									
		150	5.0 V	5.0 V	GND	5.0 V		GND	GND		5.0 V			L	L	H	GND									
		151		GND	5.0 V		GND	5.0 V	GND		5.0 V	L		H												
		152		GND	5.0 V		GND	5.0 V	5.0 V		GND			L												
		153		5.0 V	GND		GND	5.0 V	GND		5.0 V															
		154			5.0 V	GND	GND	GND			GND															
		155			GND	GND	5.0 V	5.0 V																		
		156	GND			5.0 V					5.0 V			H				5.0 V								
		157	5.0 V			GND			5.0 V					L				GND								
		158	GND	GND	5.0 V				GND		GND			L				5.0 V								
		159	5.0 V				GND	GND	5.0 V		5.0 V			H	H	H	L	GND								
		160					5.0 V	5.0 V	GND		GND															
		161				5.0 V	GND	5.0 V	GND		GND															
		162				5.0 V	GND	GND	5.0 V		GND	H														
		163				GND	5.0 V		GND		5.0 V	H														
		164	GND	5.0 V	GND	GND	5.0 V		5.0 V		5.0 V	L	L	L	L	H	5.0 V									
		165	GND	5.0 V	GND	GND	GND		GND		GND	L	L	H	H	L	5.0 V									

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TABLE III. Group A inspection for device type 51 -Continued.

Symbol	MIL-STD-883 method	Cases E,F,2	Terminal conditions 1/																	Measured terminal	Test limits						Unit
			A_4	B_3	A_3	B_2	A_2	B_1	A_1	V_{SS}	C_{IN}	S_1	S_2	S_3	S_4	C_{OUT}	B_4	V_{DD}		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 (Fig. 8)	166	GND	GND	GND	GND	GND	GND	IN	GND	GND	OUT					GND	5.0 V	A_1 to S_1		2250		3375		2250	ns
		167				GND	GND	IN	GND			OUT							B_1 to S_1							
		168				GND	IN	GND					OUT						A_2 to S_2							
		169				IN	GND						OUT						B_2 to S_2							
		170			IN	GND								OUT					A_3 to S_3							
		171		IN	GND									OUT					B_3 to S_3							
		172	IN	GND											OUT				A_4 to S_4							
		173	GND												OUT		IN		B_4 to S_4							
t_{PHL2}		174							5.0 V		IN	OUT					GND		C_{IN} to S_1							
		175							5.0 V				OUT						C_{IN} to S_2							
		176					5.0 V		5.0 V					OUT					C_{IN} to S_3							
		177			5.0 V		5.0 V		5.0 V						OUT				C_{IN} to S_4							
t_{PHL3}		178	5.0 V		GND		GND		GND		GND					OUT	IN		B_4 to C_{OUT}	750		750		750		
t_{PHL3}		179	IN		GND		GND		GND		GND					OUT	5.0 V		A_4 to C_{OUT}	750		750		750		
t_{PHL4}		180	5.0 V		5.0 V		5.0 V	GND	5.0 V		IN					OUT	GND		C_{IN} to C_{OUT}	300		450		300		
t_{PHL4}		181	GND	5.0 V	GND	5.0 V	GND	5.0 V	GND		IN					OUT	5.0 V		C_{IN} to C_{OUT}	300		450		300		
t_{PLH1}		182		GND		GND	GND	GND	IN		GND	OUT					GND		A_1 to S_1	2900		4350		2900		
		183				GND	GND	IN	GND			OUT							B_1 to S_1							
		184				GND	IN	GND					OUT						A_2 to S_2							
		185				IN	GND						OUT						B_2 to S_2							
		186			IN	GND								OUT					A_3 to S_3							
		187		IN	GND									OUT					B_3 to S_3							
		188	IN	GND											OUT				A_4 to S_4							
		189	GND												OUT		IN		B_4 to S_4							
t_{PLH2}		190							5.0 V		IN	OUT					GND		C_{IN} to S_1							
		191							5.0 V				OUT						C_{IN} to S_2							
		192					5.0 V		5.0 V					OUT					C_{IN} to S_3							
		193			5.0 V		5.0 V		5.0 V						OUT				C_{IN} to S_4							
t_{PLH3}		194	5.0 V		GND		GND		GND		GND					OUT	IN		B_4 to C_{OUT}	750		1125		750		
t_{PLH3}		195	IN		GND		GND		GND		GND					OUT	5.0 V		A_4 to C_{OUT}	750		1125		750		
t_{PLH4}		196	5.0 V		5.0 V		5.0 V		5.0 V		IN					OUT	GND		C_{IN} to C_{OUT}	300		450		300		
t_{PLH4}		197	GND	5.0 V	GND	5.0 V	GND	5.0 V	GND		IN					OUT	5.0 V		C_{IN} to C_{OUT}	300		450		300		
t_{THL}	3004 (Fig. 8)	198		GND	GND	GND	GND	GND	IN		GND	OUT					GND		S_1	1000		15000		10000		
		199		GND	GND	IN	5.0 V	5.0 V	5.0 V				OUT				GND		S_2							
		200		5.0 V	IN	5.0 V	5.0 V	GND	GND					CUT			GND		S_3							
		201	IN	5.0 V	5.0 V	GND	GND		GND						OUT		5.0 V		S_4							
		202	IN	GND	GND	GND	GND		GND							OUT	5.0 V		C_{OUT}	520		780		520		
		203	5.0 V	GND	5.0 V	GND	5.0 V		5.0 V		IN					OUT	GND		C_{OUT}	520		780		520		
t_{TLH}		204	GND	GND	GND	GND	GND		IN		GND	OUT							S_1	10000		15000		10000		
		205	GND	GND	GND	IN	5.0 V	5.0 V	5.0 V				OUT						S_2							
		206	GND	5.0 V	IN	5.0 V	5.0 V	GND	GND					OUT					S_3							
		207	IN	5.0 V	5.0 V	GND	GND		GND						OUT		5.0 V		S_4							
		208	IN	GND	GND	GND	GND		GND							OUT	5.0 V		C_{OUT}	520		780		520		
		209	5.0 V	GND	5.0 V	GND	5.0 V		5.0 V		IN					OUT	GND		C_{OUT}	520		780		520		

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1/ Pins not designated may be high level logic, low level logic, or open. Exceptions are as follows: $V_{IC(POS)}$ tests, the V_{SS} terminal shall be open; $V_{IC(NEG)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.
2/ The I_{SS} test measurements shall be performed in sequence.
3/ The following timing sequence and input/output conditions shall apply:

Inputs				Outputs	
A_N	B_N	C_{IN}	$*C_{INT}$	C_{OUT}	Sum
L	L	L	L	L	L
H	L	L	L	L	H
L	H	L	L	L	H
H	H	L	H	L(H**)	L
L	L	H	L	L	H
H	L	H	H	L(H**)	L
L	H	H	H	L(H**)	L
H	H	H	H	L(H**)	H

*Internal carry
**C_{OUT} = H when N = 4

Input/output conditions

Test	V_{DD}	Input levels		Output levels	
		1	0	1	0
	Volts			Min	Max
V_{IH1}	5.0	V_{DD} 3.5 V	1.5 V V_{SS}	4.5 V	0.5 V
V_{IL1}					
V_{IH2}	10.0	V_{DD} 7.0 V	3.0 V V_{SS}	9.0 V	1.0 V
V_{IL2}					
V_{IH3}	15.0	V_{DD} 11.0 V	4.0 V V_{SS}	13.5 V	1.5 V
V_{IL3}					

4/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for individual input or measure all inputs together.
5/ See 4.4.1c.
6/ The truth table tests shall be performed in sequence.
7/ The truth table tests shall be performed at V_{IH} and V_{DD} $\leq$5 Vdc and $\geq$18 Vdc. L = V_{SS} + 0.50 V maximum and H = V_{DD} - 0.50 Vdc.

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2. Dynamic tests (test condition D) use circuit shown on figure 7, or equivalent. T_A shall be 125°C minimum. Test duration shall be in accordance with table I method 1015.

c. Interim and final electrical test parameters shall be as specified in table II herein.
d. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter measurements.

4.2.1 Percent defective allowable (PDA).

a. The class S devices PDA shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.
b. Static burn-in I and II failures shall be cumulative for determining PDA.
c. The class B devices PDA shall be in accordance with MIL-M-38510. Dynamic burn-in is not required.
d. Those devices whose measured characteristics after burn-in exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed to the burn-in lot. The verified failures divided by the total number of devices in the lot initially submitted to burn-in shall be used to determine the percent defective for the lot, and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.3.1 Qualification extension. When authorized by the qualifying activity, for qualification inspection, if a manufacturer qualifies to a 51 device type which is manufactured identically to a 01 device type on this specification. Then the 01 device type may be part I qualified by conducting only group A electrical tests and any electricals specified as additional group C subgroups and submitting data in accordance with MIL-M-38510, appendix (i.e., groups A, B, C, D, and E tests are not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.
b. Subgroups 5 and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.
c. Subgroup 4 (Ci measurements) shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S steady state life (accelerated) test circuits shall be submitted to the qualifying activity for approval. When the alternate steady state life test is used, the circuit on figure 3, or equivalent, shall be used.
b. A special subgroup shall be added using an LTPD of 15 for class S and B , and shall be performed on each inspection lot for device types 01, and measured only for initial qualification and after process or design changes for device types 51. This subgroup shall consist of a high voltage test of the input protection circuits, V_{ZAP} (see 4.5.3).

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c. End point electrical parameters shall be as specified in table II herein, and shall consist of those subgroups specified in table IIa of test method 5005 of MIL-STD-883, and table II herein also. Delta limits shall apply only to subgroup 5 of group B inspections, and shall consist of tests specified in table IV herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection, and shall consist of tests specified in table IV herein.
b. Steady state life test (method 1005 of MIL-STD-883) conditions:
   1. Test condition D, as specified in 4.5.2, and as shown on figure 5, or equivalent.
   2. T_A = 125°C, minimum.
   3. Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardened (see 3.6.1). When group E testing is performed it shall be in accordance with table V of method 5005 of MIL-STD-883 and 4.5.5 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit V_{SS} terminal, unless otherwise specified. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cooldown procedures. When these tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to a temperature of 25° +3°C, and the electrical parameter end-point measurements shall then be performed.

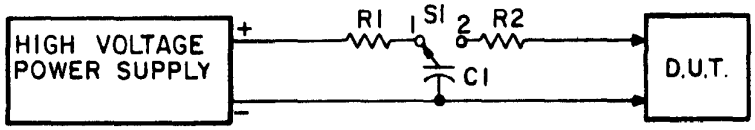
TABLE IV. Delta limits at 25°C.

Parameter	Device type	
	01	51
I_{SS}	±125 nA	±125 nA
V_{OL1}	±.04 V	---
V_{OH1}	±.08 V	---
I_{OL1}	---	±15 percent
I_{OH1}	---	±15 percent

NOTE: Each of the above parameters shall be recorded before and after the required burn-in and life tests to determine delta's (Δ).

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4.5.3 High voltage (V_{ZAP}) test of input protection circuits. All input terminals (up to a maximum of four) of the DUT shall be subjected to a voltage pulse from a 100 pF source charged to 400 volts. This destructive test shall be conducted as follows using the test circuit on figure 7.



NOTES:
1. V_{ZAP} = 400 V charge on C1.
2. 1 MΩ ≤ R1 ≤ 50 MΩ.
3. R2 = 1.5 kΩ.
4. C1 = 100 pF.
5. S1 = Hg-wetted "bounceless" relay.

FIGURE 7. High voltage (V_{ZAP}) test circuit.

a. Measure I_{IL} and I_{IH} at the inputs selected, as stated above, at 25°C. The test limit for each input tested shall be ±10 nA at the specified V_{DD}. Measure I_{SS} on the DUT at 25°C. The test limit for this measurement shall be increased a maximum of 20 percent of the specified I_{SS} table III limit at the specified V_{DD}.
b. V_{ZAP} is applied to the DUT in the following modes (see table V) by charging C1 to V_{ZAP} with S1 in position 1 and then switching to position 2.

TABLE V. Modes for high-voltage test.

Mode	+ Terminal	- Terminal
1	V_{DD}	Input
2	Input	V_{SS}
3	Input	Associated output

c. Within 24 hours repeat the I_{SS}, I_{IH}, and I_{IL} measurements on the same terminals as performed above. If the DUT exhibits leakage currents in excess of the specified limits after the V_{ZAP} test, it shall be classified as a failure.

4.5.4 Quiescent supply current (I_{SS}) test. When performing quiescent supply current measurements (I_{SS}), the meter shall be placed so that all currents flow through the meter.

4.5.5 Radiation hardness assurance (RHA) testing. The RHA testing shall be performed in accordance with test procedures and sampling specified in table V of method 5005 of MIL-STD-883 and herein:

a. Before irradiation, selected samples shall be assembled in qualified packages and pass the governing electrical parameters (group A subgroup 1 at 25°C) and also be subjected to the threshold-voltage test in table VIII in order to calculate the delta threshold (ΔV_T) after irradiation.
b. The devices shall be subjected to a total radiation dose as specified in MIL-M-38510 for the radiation hardness assurance (RHA) level being tested and meet the end point electrical parameters as defined in table VI at 25°C, after exposure. The start and completion of the end point electrical parameter measurements shall not exceed 2 hours following irradiation.

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c. Threshold-voltage test circuit conditions shall be as specified in table VIII and figure 7. In situ and remote testing, the tests shall be performed with the devices biased in accordance with table VII and bias may be interrupted for up to 1 minute to remove devices to the remote bias fixture.
d. After irradiation, the devices shall pass the truth table test as specified in subgroup 7 in table III or if subgroup 7 is not required, then an equivalent truth table test shall be performed.

TABLE VI. Radiation hardness end-point electrical parameters at 25°C.

Parameter	Test limits (all device types)	V_{DD}	
		Device 01	Device 51
V_{TN}	0.3 V min	10 V dc	10 V dc
V_{TP}	2.8 V max	10 V dc	10 V dc
ΔV_T	1.4 V max	10 V dc	10 V dc
I_{SS}	100 x max limit	15 V dc	18 V dc
t_{PLH}	1.35 x max limit	5 V dc	5 V dc
t_{PHL}	1.35 x max limit	5 V dc	5 V dc

TABLE VII. Bias during exposure to radiation.

Device type	Pin connections 1/		
	10 V dc through a 30 kΩ to 60 kΩ resistor	V_{SS} = GND	V_{DD} = 10 V dc
01 and 51	1,2,3,4,5,6,7,9,15	8	16

1/ Pins not designated are open or connected to 10 V dc through a 30 kΩ to 60 kΩ resistor.

4.6 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, shall be supplied:

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and steady state life tests.
b. A copy of each radiograph.
c. The quality conformance inspection data (see 4.4).
d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).
e. Final electrical parameter data (see 4.2c).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).
b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
c. Requirements for certificate of compliance, if applicable.

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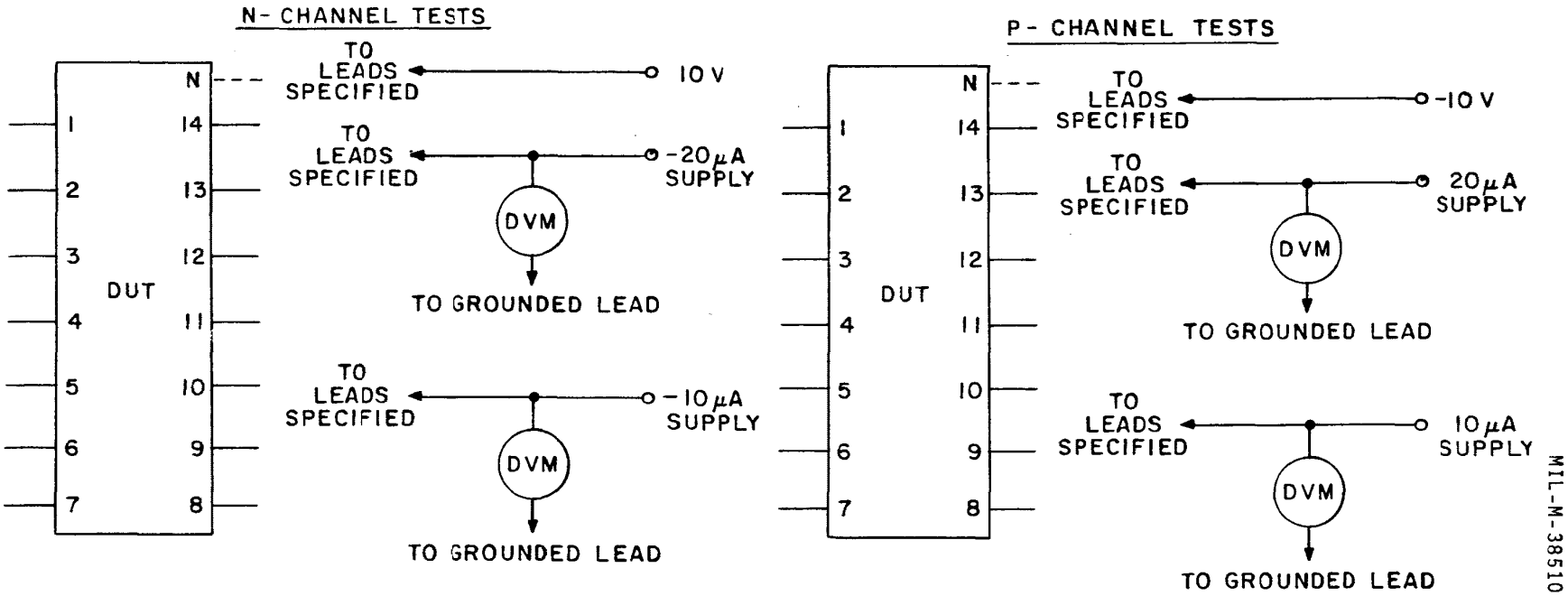


FIGURE 7. Threshold-voltage test current.

TABLE VIII. Threshhold-voltage test circuit conditions.

Device type	GND	10 V	V_{IN} measured at		GND	-10 V	V_{TP} measured at	
			-20 μA supply	-10 μA supply			20 μA supply	10 μA supply
01 51	9	2, 4, 6, 15, 16	1,3,5,7,8		9	1,3,5,7,8	2, 4, 6, 15, 16	

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d. Requirements for notification of change of product or process to procuring activity in addition to notification to the qualifying activity, if applicable.
e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.
f. Requirements for product assurance options.
g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.
h. Requirements for "JAN" marking.
i. Requirements for total dose radiation testing (see 3.6.1 and 4.5.5), if applicable.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

C_i	Input terminal-to-V_{SS} capacitance.
C_{IN}	Carry-in count from preceding stage.
C_{INT}	Internal count carry.
C_{OUT}	Carry-out count to next stage.
GND	Ground. Zero voltage potential.
T_A	Free air temperature.
V_{ZAP}	Input test voltage.
V_{IC}	Input clamp voltage.
V_{DD}	Positive supply voltage.
V_{SS}	Negative supply voltage.
I_{SS}	Quiescent supply current.

6.4 Logistic support. Unless otherwise specified, microcircuits procured for Government logistic support shall be procured to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), and lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	4008A
51	4008B

6.6 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protection devices have been designed in the chip to minimize the effect of this static buildup. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surfaces.
b. Ground test equipment and tools.
c. Do not handle devices by the leads.
d. Store devices in conductive foam or carriers.
e. Avoid use of plastic, rubber, or silk in MOS areas.
f. Maintain relative humidity above 50 percent, if practical.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - AR, SM
Navy - AS, OS, CG, MC, SH

Agent:
DLA - ES

Preparing activity:
NASA - NA

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